

Analysis of Scientific Data



Lecture 8

Modelling Association

Poll Question

In Applied Class 5, we did the Ruler drop test to collect data on reaction time. Two measurements (**Reaction1** and **Reaction2**) were taken for each subject. We wanted to compare these to see if there had been a learning effect.

Why do we not compare **Reaction1** and **Reaction2** using a two-sample t -test?

1. The distributions are not Normal
2. The observations are not independent
3. The standard deviations are not equal
4. The sample size is too large for a t -test

paired t -test

Some Recurring Themes

Many of our test statistics are of the form

$$\text{standardise} \quad \frac{\overset{\bar{x}}{\text{estimate}} - \overset{0}{\text{hypothesised value}}}{\text{se}(\text{estimate}) \quad \frac{s}{\sqrt{n}}}$$

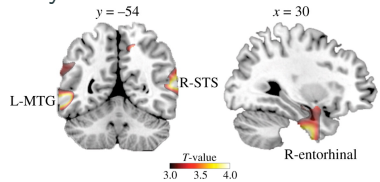
“How far off is our estimate from our hypothesised value, *relative to the underlying variability of our estimate?*”

We compare this to either a T distribution (quantitative responses) or a Normal distribution (categorical responses – Module 11).

Facebook Friends

Researchers at University College London took a sample of 125 student volunteers and used MRI to measure **grey matter (GM) density** within hundreds of small volumes of the brain. They then looked at the correlation between these densities and the **number of Facebook friends** to identify regions where GM density might be associated with the size of online social networks.

Do you have any concerns about this study?



Paper DOI: 10.1098/rspb.2011.1959

Facebook Friends – Experiment 2

The researchers took a **new sample** of 40 student volunteers and used MRI to measure GM density within the **three** small volumes of the brain that showed the strongest correlation in the first experiment. They again looked at the **correlation** between these densities and the number of Facebook friends.

The data gives standardised GM densities for one of these regions, together with the number of Facebook friends. Is there evidence of an association between these variables?

Statistical Model

- How do we model the relationship between GM density and Facebook friends?
- How do we estimate residual variability?
- How do we test for an association?
- How do we calculate the p -value?

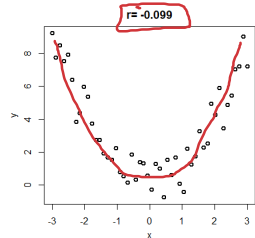
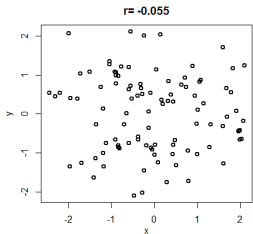
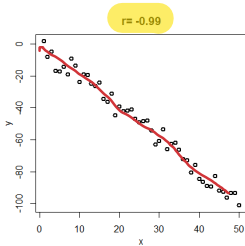
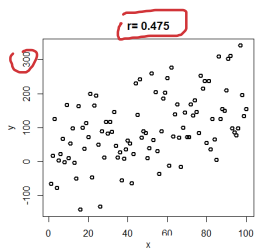
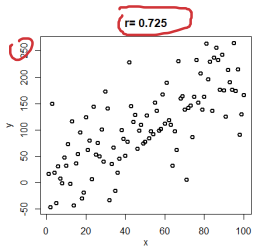
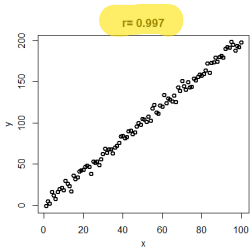
Correlation



The **correlation coefficient** measures the relative strength and direction of the **linear** relationship between two random variables.

ρ – population correlation coefficient

r – sample correlation coefficient



Poll Question

Let ρ be the correlation coefficient for the relationship between GM density and Facebook friends in the population.

We want to test

1. $H_0 : \rho = 0$ vs $H_1 : \rho > 0$
2. $H_0 : \rho = 0$ vs $H_1 : \rho < 0$
3. $H_0 : \rho = 0$ vs $H_1 : \rho \neq 0$

Poll Question

What is the sample correlation coefficient for the relationship between GM density and Facebook friends?

1. -0.1635

2. $+0.1870$

3. $+0.3808$

4. $+0.6980$

Testing $H_0 : \rho = 0$

Let ρ = population correlation coefficient between GM and Friends

Test $H_0: \rho = 0$ vs $H_1: \rho \neq 0$

Observed $r = 0.3808123$, $n = 40$

one-sided P-value = $P(R \geq 0.3808123 \mid H_0 \text{ true})$

$$= P\left(T \geq \frac{0.3808123 - 0}{\text{se}(R)}\right)$$

$$= P(T_{38} \geq 2.539)$$

$$= 1 - \text{pt}(2.539, df=38)$$

$$= 0.00767$$

$$\text{two-sided P-value} = 2 \times 0.00767 = 0.01534$$

$$\text{se}(R) = \sqrt{\frac{1-R^2}{n-2}} \\ \approx 0.15$$

Poll Question

With 40 observations of 2 variables, the degrees of freedom for the Pearson correlation coefficient are

- 1
- 2
- 37
- 38
- 39
- 40

$n-2$

$$CI = \underset{r}{\text{estimate}} \pm \underbrace{Z^{*}(0.975) \times se(\text{estimate})}$$

Z -transformation

$$Z = \frac{1}{2} \log \left(\frac{1+r}{1-r} \right) = \tanh^{-1}(r)$$

$$\sim \text{Normal} \left(\frac{1}{2} \log \left(\frac{1+\rho}{1-\rho} \right), \frac{1}{n-3} \right)$$

$$\text{for } Z: CI = 0.401 \pm 1.96 \times 0.1644$$

$$= (0.0788, 0.7232)$$

$$\text{for } \rho: CI = (\tanh(0.0788), \tanh(0.7232)) = (0.0786, 0.6189)$$

Inference for Correlation

Hypothesis test

Test $H_0 : \rho = 0$ vs $H_1 : \rho \neq 0$.

The test statistic is $t = \frac{r-\rho}{\text{se}(r)}$ where $\text{se}(r) = \sqrt{\frac{1-r^2}{n-2}}$.

The p -value is $2 \times P(T_{n-2} \geq t)$.

Confidence interval

An approximate 95% confidence interval for ρ is

$$(\tanh(z_L), \tanh(z_U))$$

where

$$(z_L, z_U) = \frac{1}{2} \log \left(\frac{1+r}{1-r} \right) \pm z_{0.975}^* \times \frac{1}{\sqrt{n-3}}.$$

Simple linear regression

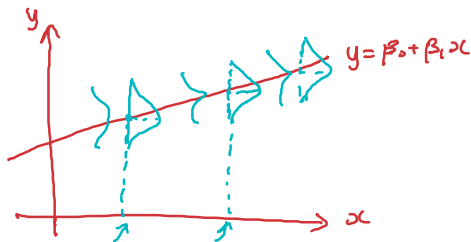
- Data: $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$

- Model:

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad i = 1, 2, \dots, n.$$

intercept *slope*

- Assumption: $\epsilon_1, \epsilon_2, \dots, \epsilon_n \stackrel{\text{iid}}{\sim} \text{Normal}(0, \sigma)$.



$$Y \sim \text{Normal}(\beta_0 + \beta_1 x, \sigma)$$

Poll Question

$$Y \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma)$$

Which of the following is **not** an assumption for linear regression?

1. Linear relationship ✓
2. Equal group sizes ✗
3. Normally distributed errors ✓
4. Constant variance of errors ✓

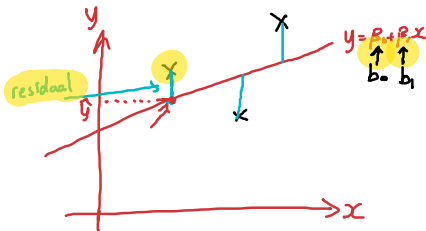
independence of errors ✓

Estimated model $y = b_0 + b_1 x$

Estimated y value $\hat{y} = b_0 + b_1 x$

Residual $\hat{e} = \hat{y} - y$

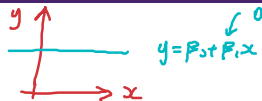
residuals = estimated errors



The best-fitting line
minimises the sum of squared
residuals $\sum_{i=1}^n \hat{e}_i^2$

Poll Question

If there is no relationship between x and y



Let β_1 be the slope coefficient for the linear relationship between GM density and Facebook friends in the population.

We want to test

1. $H_0 : \beta_1 = 0$ vs $H_1 : \beta_1 > 0$
2. $H_0 : \beta_1 = 0$ vs $H_1 : \beta_1 < 0$
3. $H_0 : \beta_1 = 0$ vs $H_1 : \beta_1 \neq 0$

Testing $H_0 : \beta_1 = 0$

Test $H_0 : \beta_1 = 0$ vs $H_1 : \beta_1 \neq 0$

Observe $b_1 = 0.002886$

one-sided P-value = $P(\beta_1 \geq 0.002886 \mid H_0 \text{ true})$

$$= P(T_{n-2} \geq \frac{0.002886 - 0}{se(b_1)})$$

$\swarrow 0.001137$

$$= P(T_{38} \geq 2.538)$$

$$= 0.0077$$

Two-sided P-value = $2 \times 0.0077 = 0.015$
same as the correlation test



There is moderate evidence of an association between
GM density and number of Facebook friends

R^2 tells us the proportion of variability explained by the model

$$\begin{aligned} 95\% \text{ CI for } \beta_1: \quad & b_1 \pm t_{n-2}^{0.975} \times \text{se}(b_1) \\ &= 0.002886 \pm 2.0244 \times 0.001137 \\ &= (0.000584, 0.005188) \end{aligned}$$

CI for y

- ① Confidence interval: plausible values for the mean
- ② Prediction interval: plausible values for a new individual observation